

Img2CADSeq: Image-to-CAD Generation via Sequence-Based Diffusion

SHIYU TAN, School of Software and BNRist, Tsinghua University, China

ZIXUAN ZHAO[†], School of Software and BNRist, Tsinghua University, ChinaHAO GAO[†], School of Software and BNRist, Tsinghua University, ChinaZHIHENG CHEN[†], School of Software and BNRist, Tsinghua University, China

XIAOLONG YIN, School of Software and BNRist, Tsinghua University, China

ENYA SHEN*, School of Software and BNRist, Tsinghua University, China and Haihe Lab of ITAI, China

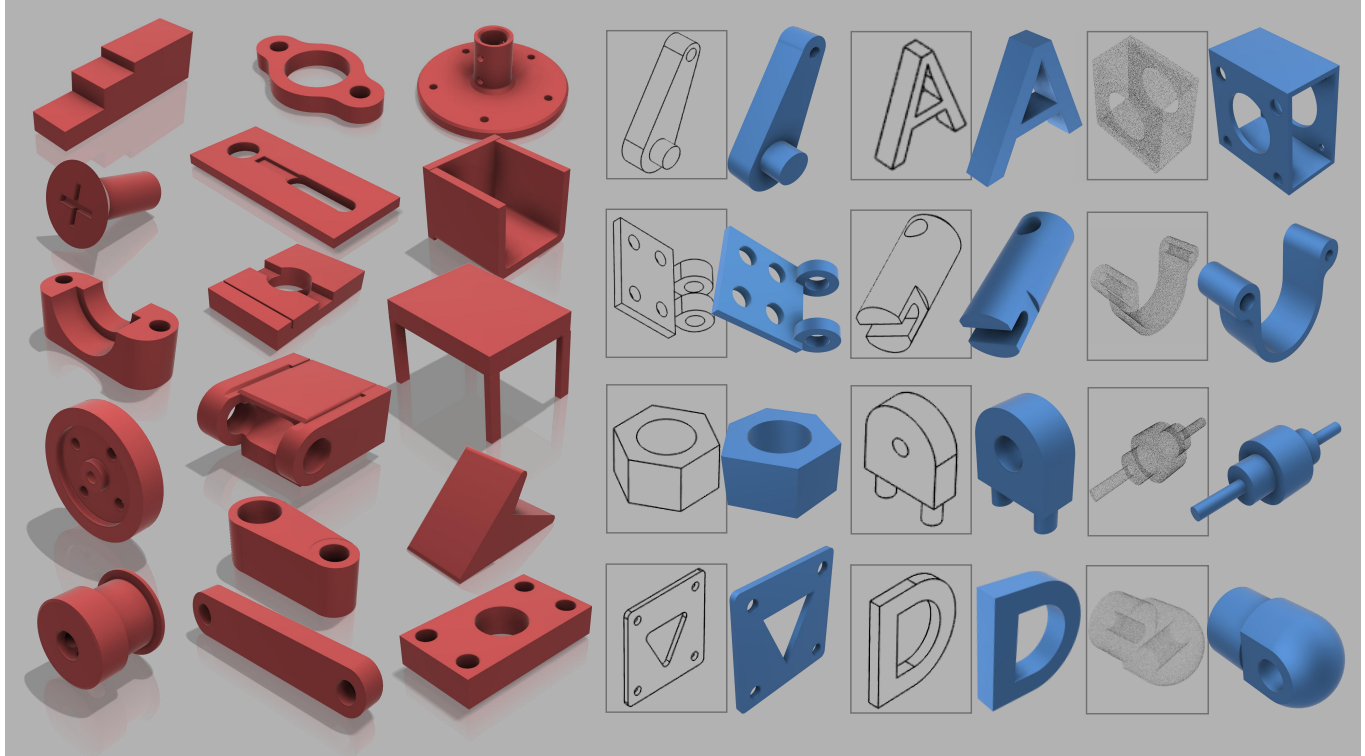


Fig. 1. Our proposed *Img2CADSeq* is a novel method based on boundary representations (BReps) structure. It is a multi-stage pipeline that can generate standardized STEP files. The fourth and fifth columns show reconstructed results generated with single-view image conditioning. The method also delivers strong results in unconditional generation, as seen in the first three columns with red parts and cloud-conditioned generation is shown in the sixth column. It surpasses existing state-of-the-art models in mechanical components generation.

[†] Authors with equal contribution.

*Corresponding author.

Authors' Contact Information: Shiyu Tan, School of Software and BNRist, Tsinghua University, Beijing, China, tansy25@mails.tsinghua.edu.cn; Zixuan Zhao, School of Software and BNRist, Tsinghua University, Beijing, China, 144946595@qq.com; Hao Gao, School of Software and BNRist, Tsinghua University, Beijing, China, hgao1743@gmail.com; Zhiheng Chen, School of Software and BNRist, Tsinghua University, Beijing, China, chenzh24@mails.tsinghua.edu.cn; Xiaolong Yin, School of Software and BNRist, Tsinghua University, Beijing, China, 1505631565@qq.com; Enya Shen, School of Software and BNRist, Tsinghua University, Beijing, China and Haihe Lab of ITAI, Beijing, China, shenenyat@tsinghua.edu.cn.



This work is licensed under a Creative Commons Attribution 4.0 International License.
SIGGRAPH Conference Papers '26, Los Angeles, CA, USA

Boundary Representation (BRep) is the standard format for Computer-Aided Design (CAD), yet reconstructing high-quality BReps from single-view images remains challenging due to the complexity of topological constraints and operation sequences. We present Img2CADSeq, a multi-stage pipeline that overcomes these limitations by encoding CAD sequences into a three-level hierarchical codebook. Guided by an importance prioritization, this strategy values profiles over details, compressing long sequences into a stable discrete latent space. To bridge the modality gap, we leverage a coarse-to-fine point cloud intermediate, aligning 2D visual features with 3D CAD sequences via contrastive learning to condition a VQ-Diffusion model. Supported by newly introduced CAD-220K and PrintCAD datasets,

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2554-8/26/07

<https://doi.org/10.1145/3799902.3811174>

our approach ensures robust industrial domain adaptation. Extensive experiments demonstrate that Img2CADSeq significantly outperforms state-of-the-art methods, producing standard STEP files that can be directly used in commercial CAD software. Code and data for this paper are at <https://github.com/Rilpraa0110/Img2CADSeq>

CCS Concepts: • **Computing methodologies** → **Shape inference**.

Additional Key Words and Phrases: Boundary representation learning, CAD program modeling, Reverse engineering

ACM Reference Format:

Shiyu Tan, Zixuan Zhao, Hao Gao, Zhiheng Chen, Xiaolong Yin, and Enya Shen. 2026. Img2CADSeq: Image-to-CAD Generation via Sequence-Based Diffusion. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers (SIGGRAPH Conference Papers '26)*, July 19–23, 2026, Los Angeles, CA, USA. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3799902.3811174>

1 Introduction

In recent years, single-view 3D generation has witnessed remarkable progress, driven by advancements in differentiable rendering techniques such as Neural Radiance Fields (NeRF) [Mildenhall et al. 2021] and Gaussian Splatting [Kerbl et al. 2023]. While these methods excel at synthesizing visually detailed 3D shapes from single-view images, the resulting representations are typically implicit fields or unstructured point splats. Crucially, they lack the clear internal topology of geometric curves and surfaces required for downstream applications [Kim and Lee 2024; Lin 2024; Rossignac 2002]. Consequently, they are ill-suited for precision-demanding tasks such as industrial design and manufacturing, raising a fundamental question: How can we generate 3D data that is not only visually coherent but also editable and compatible with engineering standards?

Boundary Representation (BRep) serves as the standard format for Computer-Aided Design (CAD) [Miyazaki et al. 2009], explicitly describing objects via precise parametric geometry and topology. However, reconstructing high-quality BRep models from single images remains a significant challenge [Kasik et al. 2005]. Unlike tensor-based formats favored by neural networks, BRep relies on complex topological relationships between geometric entities. While some studies [Wu et al. 2021] attempt to encode BReps as sequences of construction operations, these sequences are often excessively long and unstructured, making them difficult for deep models to learn effectively. To address these challenges and bridge the gap between pixel-level vision and parameter-level engineering, we present Img2CADSeq (see Fig. 1), a multi-stage framework composed of three key innovations.

First, we tackle the representation difficulty with a hierarchical encoding strategy. Inspired by importance prioritization design processes [Visser 2006], where designers favor “overall profiles” before detailing “local features”, we propose a three-layer codebook. This approach quantizes CAD operation sequences into three levels: Curve-Cluster, Sketch-Patch, and Extrude-Block. This approach naturally captures the design intent and specific parameters. Furthermore, the sequences have a sequential structure similar to natural language, which makes them easier for neural models to learn. Meanwhile, by focusing on relative geometric features rather than absolute coordinates, this structure significantly compacts the sequence length while implicitly preserving topological validity.

Second, to bridge the large semantic gap between 2D images and BRep sequences, we introduce a coarse-to-fine point cloud intermediate representation. We start by employing Dens3R [Fang et al. 2025] to lift the single-view image into an initial coarse point cloud. This approach was chosen for the flexibility of its QV linear layers, which allow for efficient training and optimization. However, a critical bottleneck is that existing single-view to point-cloud models are typically trained on non-industrial datasets (e.g., ShapeNet [Chang et al. 2015]), which fail to capture the geometric rigor of mechanical parts. To solve this, we provide two large-scale datasets: CAD-220K, a curated subset of ABC dataset comprising over 220,000 diverse 3D CAD models paired with rendered images and other modality forms. PrintCAD, a dataset of over 2,000 3D-printed components images captured under real-world lighting conditions and backgrounds. By training on these datasets, we obtain a robust coarse geometry. Subsequently, to mitigate the inherent noise in the generated geometry and capture high-frequency details, we process this output using a novel Uncertainty-Aware DGCNN (UA-DGCNN). This module leverages importance estimation to weigh reliable features, utilizing heuristic-guided resampling to extract a robust fine-grained encoding for the subsequent alignment task.

Third, we propose a robust alignment mechanism to guide generation. Merely having a point cloud is insufficient; we must semantically connect it to our CAD codebook. We employ contrastive learning to align the features of the generated point cloud with the latent space of the CAD operation sequences. This alignment ensures that the structural information from the point cloud is effectively translated into the CAD language. Finally, these features are injected as a condition into a VQ-Diffusion model [Gu et al. 2022], which generates the discrete tokens for the CAD sequence. A geometric kernel then compiles these tokens into the final BRep model (see Fig. 2).

In summary, the key contributions of this work are as follows:

- We propose a hierarchical three-layer codebook to encode CAD operation sequences. Following a “profile-to-detail” logic, this strategy compacts complex sequences into a discrete latent space suitable for diffusion modeling.
- To address the scarcity of industrial data, a key contribution of our work lies in the combination of two distinct data types: curated synthetic models (CAD-220K) and real-world captured objects (PrintCAD). By leveraging these two joint data types, we train a network to generate intermediate point clouds. This combination explicitly bridges the sim-to-real gap, enhancing the model’s generalization on mechanical parts.
- We design a novel conditioning framework that aligns 2D image-derived point clouds with CAD sequence encodings via contrastive learning. This enables our VQ-Diffusion model to predict topologically valid CAD sequences, which are subsequently compiled into standard STEP files for downstream tasks.

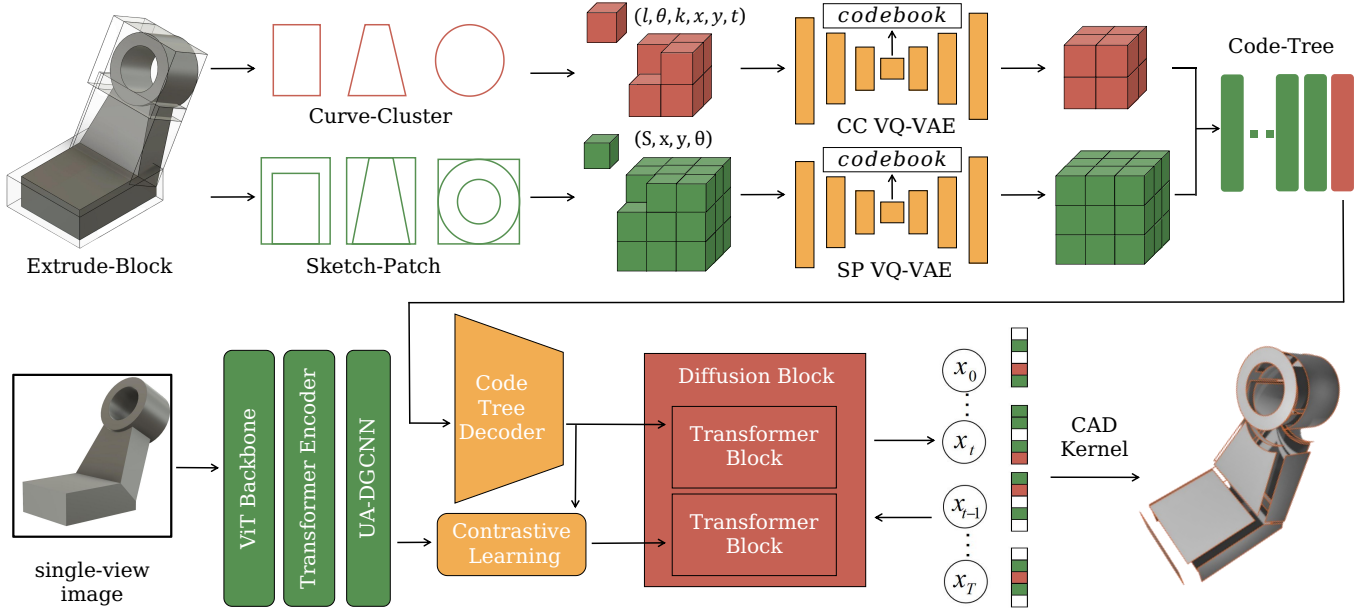


Fig. 2. **Overview of the Img2CADSeq Framework.** In the first stage, hierarchical sequence encoding represents CAD operations via a three-level codebook into a discrete space. Then we lift the input image into a 3D point cloud using a tailored network trained jointly on both synthetic and real-world data types, which is then refined by UA-DGCNN to sharpen edges and smooth surfaces. Finally, we employ contrastive learning to align the geometric embeddings with the CAD latent space, guiding a VQ-Diffusion model to predict a valid CAD operation sequence to be compiled into a watertight BRep.

2 Related Work

2.1 Parametric CAD Representation

Representation in CAD generation must bridge geometric precision and generative flexibility. Early approaches relied on standard BRep or parametric surfaces (e.g., Bézier, NURBS) [Piegl and Tiller 2012], which are geometrically exact but possess non-Euclidean structures notoriously difficult for neural networks to process. DeepCAD [Wu et al. 2021] shifted this paradigm by abstracting CAD models into sequential operations (e.g., sketch, extrude), enabling autoregressive Transformer modeling.

Recent advancements span three streams. First, direct BRep generation enforces topological consistency: BrepGPT [Li et al. 2025d] uses Voronoi Half-Patches autoregressively, while BrepDiff [Lee et al. 2025] adapts diffusion via masked UV grids. By directly generating BReps, such approaches can explore a wider topological space. Second, logic-enhanced methods like Boolean CAD [Yang et al. 2025] use Constructive Solid Geometry (CSG) [Sharma et al. 2018] to resolve complex Boolean combinations. Third, LLM-based works like CAD-Llama [Li et al. 2025c], CAD-MLLM [Xu et al. 2024], and FlexCAD [Zhang et al. 2025] convert parametric sequences into editable code scripts.

Despite these advances, methods often trade sequence learnability for topological validity. Autoregressive sequences become prohibitively long, while direct BRep generation struggles with global coherence. In contrast, we restructure the sequence-based paradigm through a coarse-to-fine lens. Our hierarchical three-level codebook compresses operations into a discrete latent space, generating fully

parametric, topologically robust models without LLM computational overhead.

2.2 Reverse Engineering

Reverse engineering geometric inputs (e.g., point clouds) into CAD models remains challenging. Traditional analytical methods [Roth and Levine 1993; Xia et al. 2020] for primitive segmentation and surface fitting demand manual intervention and lack watertight guarantees. Geometric Deep Learning [Atz et al. 2021; Bronstein et al. 2017; Cao et al. 2020] automates this. TransCAD [Dupont et al. 2024] pioneered predicting modeling sequences directly from point clouds via Transformers. AutoBRep [Xu et al. 2025] reconstructs BRep topology by learning surface adjacencies. Addressing manufacturing artifacts, DeFillet [Jiang et al. 2025] removes fillets to recover sharp edges. Recently, CAD-Recode [Rukhovich et al. 2025] utilized LLMs to decode point clouds into executable Python scripts, automating “scan-to-code”.

However, current methods bifurcate into two extremes: traditional fitting lacks editability, while LLM-based approaches are resource-intensive and noise-sensitive. Our work bridges this gap by automating “geometry-to-sequence” extraction focused on heuristic-guided resampling [Xie et al. 2025]. Specifically, our UA-DGCNN selectively resamples points to preserve sharp edges and smooth surfaces. This tailored approach effectively filters noise from intermediate geometry, enabling precise modeling sequence recovery.

2.3 Image-Driven CAD Reconstruction

Single-view image reconstruction is a prominent frontier. While neural rendering (e.g., NeRF) and diffusion-based mesh generators (Wonder3D [Long et al. 2024], TripoSR [Tochilkin et al. 2024]) produce impressive visual geometries, their outputs lack the clear internal CAD topology of curves and surfaces.

To achieve CAD generation, recent works utilize intermediate proxies or direct BRep generation. CADDreamer [Li et al. 2025b] generates normal maps via 2D diffusion priors, followed by geometric optimization to recover BReps. Img2CAD [You et al. 2025] uses Vision-Language Models (VLMs) to factorize images into SVGs, inferring 3D commands via code. Concurrently, GenCAD [Alam and Ahmed 2024] aligns visual features with CAD sequences via contrastive pretraining for diffusion generation, while CADCrafter [Chen et al. 2025] employs Diffusion Transformers (DiT) in a structured latent space. Similarly, HoLa [Liu et al. 2025] unifies geometry and topology in a holistic latent space.

However, these methods face domain adaptation and information loss bottlenecks. First, synthetic datasets like ShapeNet dominate, which lack manufacturable shapes and realistic renders. Second, approaches like CADDreamer and Img2CAD rely on 2D priors (normal maps or SVGs) to infer 3D structures, inherently losing depth and topology. This makes reconstruction ill-posed and prone to hallucinations.

To resolve the data gap, we utilize CAD-220K, a curated subset of the ABC dataset [Koch et al. 2019], alongside PrintCAD, a collection of 3D-printed solids. To bridge the modality gap, we upgrade the intermediate representation to 3D point clouds to preserve spatial structure. Crucially, rather than using them merely as inputs, we employ contrastive learning to map image features directly to the CAD sequence space. This ensures the generation of structurally aligned, STEP-standard CAD models from a single image.

3 Methodology

As illustrated in Fig. 2, the Img2CADSeq framework addresses the ill-posed single-view reconstruction problem via a structured, three-stage pipeline designed to progressively resolve geometric ambiguity.

First, Hierarchical Sequence Encoding. We introduce a three-level sequence encoder. Driven by the prior that shape recognition proceeds from “global structure to local details”, we design a hierarchical three-level codebook. Unlike flat representations, we decouple global semantics from local geometry, encoding CAD sequences into a compact, discrete latent space via Vector Quantized Variational Autoencoders (VQ-VAE) [Van Den Oord et al. 2017].

Second, Geometry-Aware Feature Extraction. To bridge the domain gap, we lift the single-view image into a 3D point cloud. To mitigate noise and redundancy, a UA-DGCNN refines this intermediate geometry, extracting robust feature embeddings that encode essential topological cues.

Third, Cross-Modal Alignment and Generation. We employ a contrastive learning framework to align the point cloud embeddings with the CAD latent space. These aligned features serve as structural conditions for a VQ-Diffusion model. Finally, a geometric kernel compiles the predicted tokens into a BRep model.

3.1 CAD Sequence Encoder

A fundamental bottleneck in CAD generation lies in the spatial-semantic coupling inherent in raw operation sequences. While sharing the goal of sequence-based generation with recent methods like SkexGen [Xu et al. 2022], we avoid parallel encoders that limit vertical hierarchical dependencies. Building upon the hierarchical VQ-VAE paradigm of HNC-CAD [Xu et al. 2023], we address its over-reliance on absolute Euclidean coordinates—a formulation that entangles local geometry with global placement and violates translation invariance, thus inhibiting transferable geometric primitives.

To address this, we introduce a three-level codebook that utilizes a novel sorting mechanism and a local coordinate formulation. The modeling process is factorized into: Extrude-Block (EB) for global semantics, Sketch-Patch (SP) for topological layout, and Curve-Cluster (CC) for local geometry. (see Fig. 3)

The EB layer abstracts modeling operations into parametric primitives. We encode the construction parameters into a global vector $\mathbf{e}^{\text{eb}} \in \mathbb{R}^{512}$ via a Multi-Layer Perceptron (MLP):

$$\mathbf{e}^{\text{eb}} = \text{MLP}_{\text{global}}(\mathbf{n}_{\text{sketch}} \oplus \mathbf{p}_{\text{origin}} \oplus h_{\text{ext}} \oplus b_{\text{type}}), \quad (1)$$

where $\oplus$ denotes concatenation, $\mathbf{n}_{\text{sketch}}, \mathbf{p}_{\text{origin}} \in \mathbb{R}^3$ are the unit normal vector and origin of the sketch plane, h_{ext} is extrusion depth, and $b_{\text{type}} \in \{0, 1\}^3$ is a one-hot Boolean indicator (New, Join, Cut). The 10-dimensional raw feature is projected to the latent space $\mathbb{R}^{512}$.

The SP layer simulates structural prioritization and establishes a spatial anchor for the relative Curve-Cluster nodes. Unlike sorting in ascending order, we sort m -th loop using a score: $S_m = \underbrace{(w_m \times h_m)}_{\text{Area}} + \alpha \cdot \underbrace{\sqrt{w_m^2 + h_m^2}}_{\text{Diagonal}} - \beta \cdot \underbrace{(x_m + y_m)}_{\text{Position Penalty}}$. With $\alpha \gg \beta$, dominant profiles (large area or diagonal scale) rank first, using position solely as a tie-breaker based on top-left coordinates (x_m, y_m) .

To enable shift-invariant encoding for the subsequent CC layer, the SP level establishes a spatial anchor:

$$\mathbf{e}_m^{\text{sp}} = \text{MLP}_{\text{layout}}(S_m \oplus \mathbf{p}_{\text{start}}^{(m)} \oplus \theta_{\text{start}}^{(m)}), \quad (2)$$

where $\mathbf{p}_{\text{start}}^{(m)} = (x_0, y_0)$ and $\theta_{\text{start}}^{(m)}$ denote the absolute starting coordinates and tangent angle of the first curve in Loop L_m .

The CC layer discards absolute coordinates in favor of a local Frenet-frame encoding. The i -th primitive is formulated as an 8-dimensional feature $\mathbf{f}_i$ by concatenating geometric parameters with a type indicator and mapped to a latent code:

$$\mathbf{e}_i^{\text{cc}} = \text{MLP}_{\text{curve}}(\mathbf{f}_i), \quad \text{where } \mathbf{f}_i = [l_i, \Delta\theta_i, \kappa_i, \delta x_i, \delta y_i] \oplus \mathbf{t}_i. \quad (3)$$

Geometric parameters include chord length l_i , relative tangential deviation $\Delta\theta_i$, and curvature κ_i . We also explicitly model residual offsets $(\delta x_i, \delta y_i)$ to better support geometric closure. $\mathbf{t}_i$ is a one-hot vector categorizing the primitive into {Line, Arc/Circle, EOS}. By delegating loop segmentation to the SP level, we eliminate redundant <SEP> tokens, allowing the CC sequence to form a continuous trajectory terminated solely by <EOS> to improve sequence learnability.

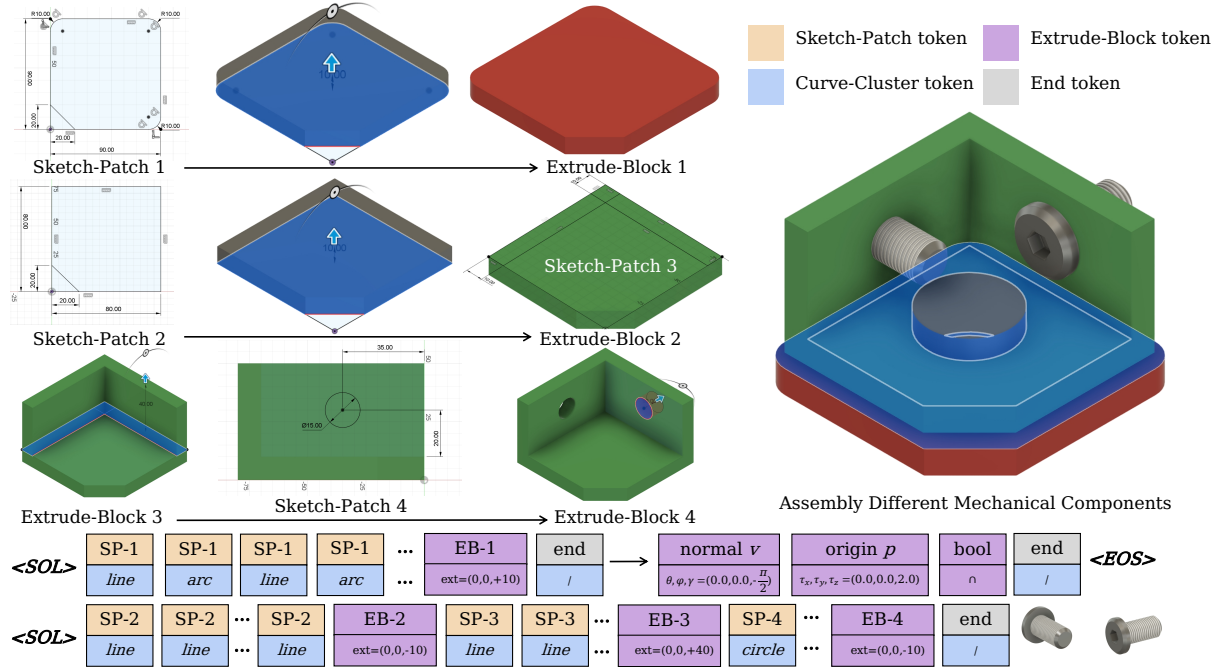


Fig. 3. **Workflow of Hierarchical Entity Construction.** At the base level, the **Curve-Cluster** parameterizes geometric primitives, which form closed loops in the **Sketch-Patch**. These loops are then lifted into 3D space via a normal vector and origin to perform extrusion and Boolean operations, resulting in an **Extrude-Block**. Multiple blocks are finally assembled to yield the target solid. This process mirrors the construction history of standard CAD workflows, preserving human design intent.

To learn the discrete latent codes, we employ three independent Vector Quantized Variational Autoencoders (VQ-VAE) corresponding to the EB, SP, and CC levels. Each VQ-VAE adopts a symmetric Transformer-based architecture. Level-specific MLPs are utilized to project raw heterogeneous features into a unified latent space, and a Masked Skip Connection strategy is applied between the encoder E_ϕ and decoder D_ψ to strictly enforce codebook reliance. The networks are trained by optimizing a total loss $\mathcal{L}$, which is formulated as a weighted sum of reconstruction, quantization, and geometric closure terms:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{vq}} + \lambda_{\text{cls}} \mathcal{L}_{\text{closure}}. \quad (4)$$

Specifically, the hybrid reconstruction loss $\mathcal{L}_{\text{recon}}$ (balanced by $\alpha = 1.0, \beta = 0.5$) accounts for heterogeneous data types by combining a Mean Squared Error (MSE) term for continuous geometric parameters (such as $l_i, \Delta\theta_i$ in CC or h_{ext} in EB) and a Cross-Entropy (CE) term for discrete categorical indicators (such as $\mathbf{t}_i$ or b_{type}). The quantization loss $\mathcal{L}_{\text{vq}} = \|\text{sg}[\mathbf{z}_e] - \mathbf{z}_q\|_2^2 + 0.25\|\mathbf{z}_e - \text{sg}[\mathbf{z}_q]\|_2^2$ employs a standard commitment mechanism to stabilize codebook updates, where $\text{sg}[\cdot]$ denotes the stop-gradient operator. Finally, to enforce the watertightness of the generated CAD models, we introduce a loop closure regularization $\mathcal{L}_{\text{closure}}$ specifically for the CC level, which penalizes the cumulative Euclidean error of the relative path reconstruction, defined as $\|\sum \mathcal{T}(l_i, \Delta\theta_i)\|_2^2$, where $\mathcal{T}$ transforms the relative intrinsic parameters back to global displacement vectors, to minimize open-loop artifacts. We select $\lambda_{\text{cls}} = 1.0$ in the experiment.

3.2 Point Cloud Acquisition

To lift the 2D input $I \in \mathbb{R}^{H \times W \times 3}$ into a 3D point cloud, we adapt the Dens3R method. To bridge the gap between generic objects and rigid industrial parts, we employ Parameter-Efficient Fine-Tuning (PEFT) on the Query and Value linear layers. We optimize projection weights Θ to align spatial attention with CAD priors:

$$\Theta^* = \arg \min_{\Theta} \mathcal{L}_{\text{rec}}(f(I; \mathbf{W}_{\text{frozen}}, \Theta), \mathcal{P}_{\text{GT}}), \quad (5)$$

where $\mathbf{W}_{\text{frozen}}$ represents the frozen weights of the backbone, Θ denotes the learnable parameters of the network f , and $\mathcal{P}_{\text{GT}}$ refers to the corresponding ground truth 3D point cloud. By refining Θ , we effectively recalibrate the reasoning of the model to align with the rigid geometric priors of CAD models, generating an initial coarse point cloud $\mathcal{P}_{\text{raw}}$ with reasonable global structures but noisy boundaries.

To refine noisy boundaries without losing sharp edges, we propose an Uncertainty-Aware DGCNN. The backbone processes 3D vertices via four sequential EdgeConv layers to predict a geometric importance score $s_i \in [0, 1]$. A high s_i indicates regions of high curvature or geometric importance (e.g., sharp edges, corners). To transform these scores into a selection mask, we design a Heuristic-guided Resampling strategy that balances feature preservation with global coverage. The probability $p(p_i)$ of selecting a point p_i is

formulated as a mixture distribution:

$$p(p_i) = \underbrace{\lambda \cdot \frac{e^{\beta s_i}}{\sum_k e^{\beta s_k}}}_{\text{Saliency Term}} + \underbrace{(1 - \lambda) \cdot \frac{1}{N}}_{\text{Coverage Term}}. \quad (6)$$

Here, the Saliency Term utilizes a Boltzmann distribution (controlled by temperature β) to aggressively prioritize high-importance points, ensuring that sharp features are almost deterministically retained. The Coverage Term ensures a uniform background sampling to prevent voids in planar regions. λ serves as the trade-off coefficient.

Finally, we sample M points based on $p(p_i)$ to form the refined point cloud $\mathcal{P}_{\text{ref}}$. This set is then projected into the latent space via the geometric encoder $\mathcal{E}$ to obtain the structural embedding $\mathbf{z}_{\mathcal{P}} = \mathcal{E}(\mathcal{P}_{\text{ref}})$, which serves as the geometry-aligned condition for the subsequent CAD sequence generation.

3.3 Cloud-to-CAD Conversion

Once the geometric embedding $\mathbf{z}_{\mathcal{P}}$ is obtained from the point cloud, the critical next step is to map it into the semantic space of CAD operations. Instead of training a CAD autoencoder in isolation, we propose a joint contrastive learning framework to enforce a shared manifold between the two domains.

We formulate the CAD sequence encoder using a causal Transformer architecture. Let a CAD program $\mathcal{S}$ be represented by a sequence of discrete codebook embeddings $\mathcal{S} = (\mathbf{e}_1, \dots, \mathbf{e}_L)$, where $\mathbf{e}_i$ corresponds to the hierarchical tokens. We append learnable positional encodings to these tokens and process them through causal self-attention layers to produce the global CAD embedding $\mathbf{z}_{\mathcal{S}}$. Unlike unsupervised reconstruction, we optimize this representation to be distinct yet semantically aligned with $\mathbf{z}_{\mathcal{P}}$.

We employ the InfoNCE loss to maximize the mutual information between matched pairs while pushing apart mismatched ones. Given a batch of B synchronized pairs $\{(\mathcal{S}_i, \mathcal{P}_i)\}_{i=1}^B$, we treat the corresponding $(\mathbf{z}_{\mathcal{S}_i}, \mathbf{z}_{\mathcal{P}_i})$ as the positive pair, and the remaining $2(B - 1)$ samples within the batch as negatives. The symmetric contrastive loss is defined as:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(\mathbf{z}_{\mathcal{S}_i}, \mathbf{z}_{\mathcal{P}_i})/\tau)}{\sum_{k=1}^{2B} \mathbb{I}_{[k \neq i]} \exp(\text{sim}(\mathbf{z}_{\mathcal{S}_i}, \mathbf{z}_{\text{feat},k})/\tau)}, \quad (7)$$

where $\text{sim}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ denotes cosine similarity, τ is the learnable temperature, and $\mathbf{z}_{\text{feat}} \in \{\mathbf{z}_{\mathcal{S}}\} \cup \{\mathbf{z}_{\mathcal{P}}\}$ represents the set of all embeddings in the batch.

With the aligned condition $\mathbf{c} = \mathbf{z}_{\mathcal{P}}$ established, we frame the CAD generation as a conditional discrete diffusion process. The target CAD operation sequence is first quantized into discrete indices $\mathbf{x}_0 \in \{1, \dots, K\}^L$ via our codebook. We adopt a VQ-Diffusion framework utilizing an absorbing state transition. The forward process $q(\mathbf{x}_t | \mathbf{x}_{t-1})$ progressively corrupts the sequence by replacing tokens with a generic [MASK] token. The reverse process $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{c})$ aims to recover the clean topology $\mathbf{x}_0$ from the noisy state $\mathbf{x}_t$, explicitly guided by the structural condition $\mathbf{c}$.

We parameterize the denoising network to predict the probability distribution of the original token $\mathbf{x}_0$ directly at each timestep t :

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{c}) \propto \sum_{\tilde{\mathbf{x}}_0} q(\mathbf{x}_{t-1} | \mathbf{x}_t, \tilde{\mathbf{x}}_0) p_\theta(\tilde{\mathbf{x}}_0 | \mathbf{x}_t, \mathbf{c}), \quad (8)$$

where $p_\theta(\tilde{\mathbf{x}}_0 | \mathbf{x}_t, \mathbf{c})$ is modeled by a Transformer decoder that cross-attends to the point cloud embedding $\mathbf{c}$. The training objective combines the variational lower bound (VLB) with an auxiliary cross-entropy loss:

$$\mathcal{L}_{\text{gen}} = \mathcal{L}_{\text{vlb}} + \lambda_{\text{aux}} \mathbb{E}_{t, \mathbf{x}_t} [-\log p_\theta(\mathbf{x}_0 | \mathbf{x}_t, \mathbf{c})]. \quad (9)$$

4 Experiments

4.1 Setup

To ensure robust sequence learning and sim-to-real generalization, we curate distinct data subsets for our pipeline stages.

For the sequence learning, we train the hierarchical codebook on DeepCAD [Wu et al. 2021], a dataset comprising ground-truth CAD operation sequences. Filtering out compilation failures and non-manifold geometry yields 168,656 valid CAD programs with sequence lengths ranging from 3 to 59 operations. Our sequence encoder uses a standard 80/10/10 train/val/test split in line with existing baselines to ensure a fair comparison.

To support the point cloud lifting stage, we utilize CAD-220K (Synthetic), a curated subset of the ABC dataset [Koch et al. 2019] filtered by surface count. Observing that models with 11–50 faces constitute the vast majority (339,489 in total), we proportionally downsample the data to establish balanced complexity tiers: 40K models (1–10 faces), 120K (11–50 faces), 30K (51–100 faces), and 30K (>100 faces). For these 220K models, we generate the corresponding STLs, point clouds, and four-view rendered images.

To explore sim-to-real translation, we introduce PrintCAD, a collection of over 2,000 3D-printed solids. For each model, we systematically capture four views under real-world lighting. These objects exhibit manufacturing artifacts and texture noise, offering an evaluation of model robustness beyond synthetic renders. Notably, both CAD-220K and PrintCAD are exclusively utilized to fine-tune our image-to-point-cloud module, ensuring that the point cloud embeddings align with the CAD latent space.

For the visual rendering of our synthetic models, we use DaVinciVisualizer to synthesize photorealistic images with randomized azimuth and elevation, ensuring diverse industrial viewpoints. Point clouds ($N = 4,096$) are sampled from BReps compiled via OpenCascade. We enforce strict settings (0.001 linear deflection, 0.1 angular deflection) to first convert it into a mesh, and then use Trimesh to produce the uniform sampling of points.

4.2 Implementation Details

We implement Img2CADSeq using PyTorch and conduct all training on a server equipped with 8 NVIDIA A100 (40GB) GPUs.

The three independent VQ-VAEs (for EB, SP, and CC levels) are trained on the DeepCAD dataset for 200 epochs. We use the AdamW optimizer with a global batch size of 1,024 (128 per GPU) and a cosine learning rate schedule warmed up to 1×10^{-4} . The contrastive model for geometry-aware feature alignment is trained for 300 epochs with a global batch size of 512 (64 per GPU) and a learning rate of 1×10^{-3} . During this stage, we employ a bootstrapping data augmentation strategy that subsamples 2,048 points from dense 4,096-point clouds during each forward pass. The generative VQ-Diffusion Transformer is trained for 50,000 iterations with a global batch size of 2,048 (256 per GPU). We optimize the model using AdamW with a base learning

rate of 1×10^{-4} and a linear warmup of 5,000 steps. A linear noise scheduler is applied over 100 diffusion steps.

4.3 Metrics

We evaluate our method using a comprehensive set of standard metrics tailored to the specific objectives of each task. For image-conditioned generation, we assess geometric accuracy via Chamfer Distance (CD), evaluate structural integrity through the Ratio of Hanging Faces (HF), and measure primitive segmentation quality using Segmentation Accuracy (Seg Acc). For point cloud-conditioned generation, we quantify point-level geometry matching using Accuracy (Acc) and Completeness (Comp), while assessing primitive-level topology recovery via Precision and Recall. Finally, for unconditional generation, we evaluate distribution similarity against the reference set using Maximum Mean Discrepancy (MMD) and Jensen-Shannon Divergence (JSD). We also quantify generation diversity through Coverage (COV), Novelty (Nov), and Uniqueness (Uniq), and assess the programmatic validity of the generated sequences using the Invalid Rate (IR). Detailed definitions and implementations for all metrics are provided in the Appendix.

4.4 Baselines

Our work presents a comprehensive evaluation of BRep generation across three distinct settings, with the primary focus on single-view image conditioning. Point cloud conditioning and the unconditioning experiments serve as important extensions that demonstrate the versatility and robustness of our approach.

Table 1. Quantitative results of image-conditional generation. CD is the Chamfer Distance (multiplied by 10^2), HF is the Ratio of Hanging Faces, and Seg Acc is the Part Segmentation Accuracy. Img2CAD [You et al. 2025] specializes in furniture like chairs and cabinets, so we do not showcase the results in the figure. TripoSR and Wonder3D [Long et al. 2024] are both methods that generate meshes from single-view and TripoSR generates rather better results, so we chose it to display.

Method	CD ↓	HF ↓	Seg Acc ↑
TripoSR	9.68	32.7	40.4%
HoLa	1.82	3.3	90.6%
CADDreamer	1.36	2.4	96.1%
Img2CAD	1.37	3.0	86.3%
Wonder3D	14.26	47.2	32.4%
Ours	1.21	2.2	97.2%

Image Input. Fig. 6 visually showcases the reconstruction results from synthetic and real-world images, while Tab. 1 provides quantitative comparisons. These results demonstrate that Img2CADSeq offers significant advantages in generation quality and the topological fidelity of the reconstructed BReps.

Img2CADSeq’s key strength lies in its ability to simultaneously capture low-level geometric features and high-level shape understanding. Our method generates clean, compact BReps with sharp edges, minimizing fitting errors and reconstruction failures. In contrast, competing methods generate rather distorted reconstructions. In addition, our method employs a refined three-level codebook and

heuristic-guided resampling for sharp edges and smooth surfaces. Single-view RGB image reconstructions often produce meshes with errors such as uneven surfaces and oversmoothed boundaries. As shown in Figure 6, TripoSR [Tochilkin et al. 2024] often generates shapes that resemble simple clay models. HoLa [Liu et al. 2025] generates a rather better BRep structure, but input ambiguity leads to unreasonable results, making them unsuitable for real-world applications. Since CADDreamer [Li et al. 2025b] uses primitive fitting methods, it achieves higher precision on surfaces such as tori and cones, but its output cannot align well with the input, and it fails in intersecting relationships, sometimes producing non-watertight models. In contrast, our method demonstrates superior generation that aligns closely with the ground truth CAD models, resulting in the lowest CD, HF, and highest Seg Acc among compared approaches, confirming that our generated models are both more accurate and structurally diverse.

Table 2. Quantitative results of clean point cloud-conditioned generation. Acc Err is the Accuracy Error, and the Comp Err is the Completeness Error. Since SEDNet is an improved version of HPNet [Yan et al. 2021], we chose to show SEDNet in our figure, and include HPNet’s results in the table. Acc Err and Comp Err scores are multiplied by 10^3 .

Method	Acc Err ↓	Comp Err ↓	Precision ↑	Recall ↑
HoLa	6.59	5.05	0.944	0.920
HP+P2C	12.27	49.50	0.832	0.708
SED+P2C	7.60	47.10	0.860	0.676
Ours	6.49	4.91	0.947	0.923

Table 3. Quantitative comparison of unconditional CAD generation. MMD is the Maximum Mean Discrepancy, JSD is the Jensen-Shannon Divergence, COV is the Coverage, Nov is the Novelty, Uniq is the Uniqueness, and IR is the Invalid Ratio. JSD scores are multiplied by 10^2 .

Method	MMD ↓	Nov ↑	COV ↑	Uniq ↑	JSD ↓	IR ↓
SkexGen	1.480	93.9%	73.61%	97.8%	1.129	11.23
HNC-CAD	1.296	94.7%	73.86%	90.7%	1.484	12.09
DTGBrepGen	1.083	98.8%	74.29%	95.8%	1.284	9.57
BrepDiff	1.320	93.2%	75.21%	91.8%	1.127	11.76
HoLa	1.069	92.7%	78.87%	93.3%	1.109	17.32
Ours	0.958	97.2%	80.17%	96.4%	0.839	10.89

Point Cloud Input. Fig. 7 showcases the reconstruction results from clean and ill-scanned point clouds, while Tab. 2 provides quantitative comparisons. Visually, our method consistently produces clean, watertight BRep models that preserve sharp edges and fine geometric features. In contrast, fitting-based methods like SEDNet [Li et al. 2023]+Point2CAD[Liu et al. 2024] tend to produce more complete surfaces, and the lack of topological information prevents these surfaces from being correctly trimmed. While a major limitation of HoLa is its parametrically uneditable files, the images and generated results still cannot be properly aligned. Our method

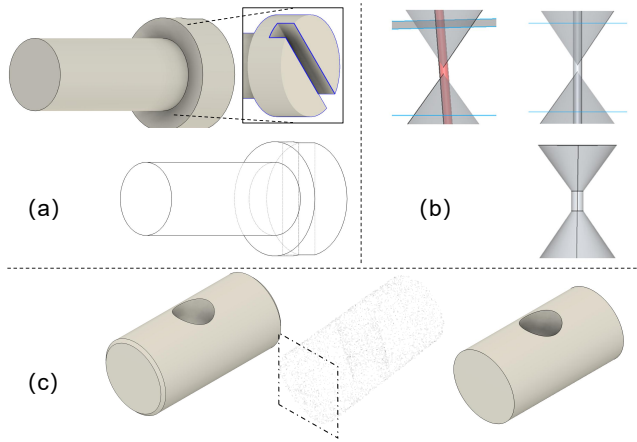


Fig. 4. **The limitations and failure cases of our work.** (a) Extreme single-view ambiguity causes plausible back-end structures but non-manufacturable in occluded areas. (b) Sequence error accumulation disrupts global geometric constraints like strict symmetry and coaxiality. (c) Limited resolution of the intermediate point cloud causes fine features to be smoothed out or omitted.

leverages a more robust codebook and noisy-friendly representation that captures both geometry and topology, supported by our method’s leading performance in Tab. 2, where it achieves the lowest Acc Err, Comp Err and the highest precision and recall scores, demonstrating its ability to cover diverse shape modes and generate a wide variety of structurally valid CAD candidates even from partial or noisy inputs.

Unconditional Generation. Fig. 8 presents a direct comparison of unconditionally generated CAD models, and Tab. 3 provides quantitative comparisons. Our method produces models that are more structurally plausible, with cleaner surfaces and more coherent mechanical features. In contrast, baseline methods such as SkexGen[Xu et al. 2022], HNC-CAD [Xu et al. 2023], and DTGBrepGen [Li et al. 2025a] often generate rather simplistic or poorly assembled shapes, while BrepDiff [Lee et al. 2025] and HoLa exhibit surface artifacts or unnatural proportions. Our results demonstrate a clear advantage in generating production-ready CAD candidates.

5 Limitations and Failure Cases

While our method advances CAD reconstruction, several limitations remain. For complex assemblies, long sequences cause accumulated errors. Additionally, while our PrintCAD dataset supports physical object reconstruction, it focuses mostly on 3D-printed materials, making it difficult to generalize to highly diverse in-the-wild scenarios. In practice, these constraints manifest in three main failure cases. First, as illustrated in Fig. 4(a), due to single-view ambiguity, the model hallucinates occluded regions based on learned priors. While topologically watertight, these generated back-end structures often lack physical and kinematic constraints; under severe occlusion, they may appear visually plausible but are physically impossible to manufacture. Second, as demonstrated in Fig. 4(b), although our codebook captures local features effectively, it lacks explicit global

constraints. Consequently, sequence error accumulation easily disrupts strict symmetry and coaxiality, and enforcing such global constraints in diffusion models remains a major challenge. Finally, as shown in Fig. 4(c), bridging the modality gap with a limited-resolution point cloud proxy (e.g., 4,096 points) often smooths out micro-structures like threads and tiny fillets during heuristic-guided resampling, causing the final CAD sequences to miss these precise manufacturing details.

6 Conclusions and Future Work

We introduced Img2CADSeq, a multi-stage pipeline that directly generates topologically valid STEP files from single views. Our hierarchical codebook and contrastive alignment—powered by the new CAD-220K and PrintCAD datasets—successfully decouple global structure from local geometry, outperforming baselines in handling the sim-to-real gap. To further resolve single-view ambiguity, future work will integrate VLMs for multimodal conditioning, enable interactive editing at intermediate stages, and explicitly enforce strict symmetry and geometric constraints. By delivering editable BReps rather than inert meshes, this work significantly advances automated reverse engineering and intelligent downstream manufacturing.

Acknowledgments

The authors thank the anonymous reviewers for their valuable feedback. This work was supported by the Shenzhen Innovation and Entrepreneurship Plan under grant numbers 20232910020 and KJZD20230923114114028.

References

- Md Ferdous Alam and Faez Ahmed. 2024. Gencad: Image-conditioned computer-aided design generation with transformer-based contrastive representation and diffusion priors. *arXiv preprint arXiv:2409.16294* (2024).
- Kenneth Atz, Francesca Grisoni, and Gisbert Schneider. 2021. Geometric deep learning on molecular representations. *Nature Machine Intelligence* 3, 12 (2021), 1023–1032.
- Michael M Bronstein, Joan Bruna, Yann LeCun, Arthur Szlam, and Pierre Vandergheynst. 2017. Geometric deep learning: going beyond euclidean data. *IEEE Signal Processing Magazine* 34, 4 (2017), 18–42.
- Wenming Cao, Zhiyue Yan, Zhiqian He, and Zhihai He. 2020. A comprehensive survey on geometric deep learning. *IEEE Access* 8 (2020), 35929–35949.
- Angel X Chang, Thomas Funkhouser, Leonidas Guibas, Pat Hanrahan, Qixing Huang, Zimo Li, Silvio Savarese, Manolis Savva, Shuran Song, Hao Su, et al. 2015. Shapenet: An information-rich 3d model repository. *arXiv preprint arXiv:1512.03012* (2015).
- Cheng Chen, Jiacheng Wei, Tianrun Chen, Chi Zhang, Xiaofeng Yang, Shangzhan Zhang, Bingchen Yang, Chuan-Sheng Foo, Guosheng Lin, Qixing Huang, et al. 2025. Cadrafter: Generating computer-aided design models from unconstrained images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 11073–11082.
- Elona Dupont, Kseniya Cherenkova, Dimitrios Mallis, Gleb Gusev, Anis Kacem, and Djamilia Aouada. 2024. Transcad: A hierarchical transformer for cad sequence inference from point clouds. In *European Conference on Computer Vision*. Springer, 19–36.
- Xianze Fang, Jingnan Gao, Zhe Wang, Zhuo Chen, Xingyu Ren, Jiangjing Lyu, Qiaomu Ren, Zhonglei Yang, Xiaokang Yang, Yichao Yan, et al. 2025. Dens3r: A foundation model for 3d geometry prediction. *arXiv preprint arXiv:2507.16290* (2025).
- Shuyang Gu, Dong Chen, Jianmin Bao, Fang Wen, Bo Zhang, Dongdong Chen, Lu Yuan, and Baining Guo. 2022. Vector quantized diffusion model for text-to-image synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10696–10706.
- Jing-En Jiang, Hanxiao Wang, Mingyang Zhao, Dong-Ming Yan, Shuangmin Chen, Shiqing Xin, Changhe Tu, and Wenping Wang. 2025. Defillet: Detection and removal of fillet regions in polygonal cad models. *ACM Transactions on Graphics (TOG)* 44, 4 (2025), 1–19.
- David J Kasik, William Buxton, and David R Ferguson. 2005. Ten CAD challenges. *IEEE Computer Graphics and Applications* 25, 2 (2005), 81–92.
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* 42, 4 (2023), 139–1.
- Hyunjeong Kim and In-Kwon Lee. 2024. Is 3dgs useful?: Comparing the effectiveness of recent reconstruction methods in vr. In *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 71–80.
- Sebastian Koch, Albert Matveev, Zhongshi Jiang, Francis Williams, Alexey Artemov, Evgeny Burnaev, Marc Alexa, Denis Zorin, and Daniele Panozzo. 2019. ABC: A Big CAD Model Dataset For Geometric Deep Learning. In *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Mingi Lee, Dongsu Zhang, Clément Jambon, and Young Min Kim. 2025. Brepdiff: Single-stage b-rep diffusion model. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers*. 1–11.
- Jing Li, Yihang Fu, and Falai Chen. 2025a. DTGBrepGen: A Novel B-rep Generative Model through Decoupling Topology and Geometry. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 21438–21447.
- Jiahao Li, Weijian Ma, Xueyang Li, Yunzhong Lou, Guichun Zhou, and Xiangdong Zhou. 2025c. CAD-Llama: leveraging large language models for computer-aided design parametric 3D model generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 18563–18573.
- Pu Li, Wenhao Zhang, Weize Quan, Biao Zhang, Peter Wonka, and Dongming Yan. 2025d. BrepGPT: Autoregressive B-rep Generation with Voronoi Half-Patch. *ACM Transactions on Graphics (TOG)* 44, 6 (2025), 1–18.
- Yuan Li, Cheng Lin, Yuan Liu, Xiaoxiao Long, Chenxu Zhang, Ningna Wang, Xin Li, Wenping Wang, and Xiaohu Guo. 2025b. CADDreamer: CAD Object Generation from Single-view Images. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 21448–21457.
- Yuanqi Li, Shun Liu, Xinran Yang, Jianwei Guo, Jie Guo, and Yanwen Guo. 2023. Surface and edge detection for primitive fitting of point clouds. In *ACM SIGGRAPH 2023 conference proceedings*. 1–10.
- Jinwei Lin. 2024. Dynamic nerf: A review. *arXiv preprint arXiv:2405.08609* (2024).
- Yujia Liu, Anton Obukhov, Jan Dirk Wegner, and Konrad Schindler. 2024. Point2CAD: Reverse Engineering CAD Models from 3D Point Clouds. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3763–3772.
- Yilin Liu, Duoteng Xu, Xingyao Yu, Xiang Xu, Daniel Cohen-Or, Hao Zhang, and Hui Huang. 2025. HOLA: B-rep generation using a holistic latent representation. *ACM Transactions on Graphics (TOG)* 44, 4 (2025), 1–25.
- Xiaoxiao Long, Yuan-Chen Guo, Cheng Lin, Yuan Liu, Zhiyang Dou, Lingjie Liu, Yuexin Ma, Song-Hai Zhang, Marc Habermann, Christian Theobalt, et al. 2024. Wonder3d: Single image to 3d using cross-domain diffusion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9970–9980.
- Ben Mildenhall, Pratul P Srinivasan, Matthew Tanicik, Jonathan T Barron, Ravi Ramamoorthi, and Ren Ng. 2021. Nerf: Representing scenes as neural radiance fields for view synthesis. *Commun. ACM* 65, 1 (2021), 99–106.
- Takashi Miyazaki, Yasuhiro Hotta, Jun Kunii, Soichi Kuriyama, and Yukimichi Tamaki. 2009. A review of dental CAD/CAM: current status and future perspectives from 20 years of experience. *Dental materials journal* 28, 1 (2009), 44–56.
- Les Piegel and Wayne Tiller. 2012. *The NURBS book*. Springer Science & Business Media.
- Jarek Rossignac. 2002. CSG-BRep duality and compression. In *Proceedings of the seventh ACM symposium on Solid modeling and applications*. 59–59.
- Gerhard Roth and Martin D Levine. 1993. Extracting geometric primitives. *CVGIP: image understanding* 58, 1 (1993), 1–22.
- Danila Rukhovich, Elona Dupont, Dimitrios Mallis, Kseniya Cherenkova, Anis Kacem, and Djamilia Aouada. 2025. Cad-recode: Reverse engineering cad code from point clouds. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 9801–9811.
- Gopal Sharma, Rishabh Goyal, Difan Liu, Evangelos Kalogerakis, and Subhransu Maji. 2018. Csgnet: Neural shape parser for constructive solid geometry. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 5515–5523.
- Dmitry Tochilkin, David Pankratz, Zexiang Liu, Zixuan Huang, Adam Letts, Yangguang Li, Ding Liang, Christian Laforte, Varun Jampani, and Yan-Pei Cao. 2024. Tripotr: Fast 3d object reconstruction from a single image. *arXiv preprint arXiv:2403.02151* (2024).
- Aaron Van Den Oord, Oriol Vinyals, et al. 2017. Neural discrete representation learning. *Advances in neural information processing systems* 30 (2017).
- Willemien Visser. 2006. Designing as construction of representations: A dynamic viewpoint in cognitive design research. *Human-Computer Interaction* 21, 1 (2006), 103–152.
- Rundi Wu, Chang Xiao, and Changxi Zheng. 2021. Deepcad: A deep generative network for computer-aided design models. in 2021 IEEE. In *CVF International Conference on Computer Vision (ICCV)*. 6772–6782.
- Shaobo Xia, Dong Chen, Ruisheng Wang, Jonathan Li, and Xinchang Zhang. 2020. Geometric primitives in LiDAR point clouds: A review. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 13 (2020), 685–707.
- Jingpeng Xie, Shiyu Tan, Yuanlei Wang, Tianle Du, Yifei Xue, and Yizhen Lao. 2025. IOVS4NeRF: Incremental Optimal View Selection for Large-Scale NeRFs. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.
- Jingwei Xu, Chenyu Wang, Zibo Zhao, Wen Liu, Yi Ma, and Shenghua Gao. 2024. Cad-mllm: Unifying multimodality-conditioned cad generation with mllm. *arXiv preprint arXiv:2411.04954* (2024).
- Xiang Xu, Pradeep Jayaraman, Joseph Lambourne, Yilin Liu, Durvesh Malpure, and Pete Meltzer. 2025. AutoBrep: Autoregressive B-Rep Generation with Unified Topology and Geometry. In *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*. 1–12.
- Xiang Xu, Pradeep Kumar Jayaraman, Joseph G Lambourne, Karl DD Willis, and Yasutaka Furukawa. 2023. Hierarchical neural coding for controllable CAD model generation. In *Proceedings of the 40th International Conference on Machine Learning*. 38443–38461.
- Xiang Xu, Karl D.D. Willis, Joseph G Lambourne, Chin-Yi Cheng, Pradeep Kumar Jayaraman, and Yasutaka Furukawa. 2022. SkexGen: Autoregressive Generation of CAD Construction Sequences with Disentangled Codebooks. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*.
- Siming Yan, Zhenpei Yang, Chongyang Ma, Haibin Huang, Etienne Vouga, and Qixing Huang. 2021. HPNet: Deep Primitive Segmentation Using Hybrid Representations. *arXiv preprint arXiv:2105.10620* (2021).
- Yingyu Yang, Xiaohong Jia, Bolun Wang, Jieyin Yang, Shiqing Xin, and Dong Ming Yan. 2025. Boolean Operation for CAD Models Using a Hybrid Representation. *Transactions on Graphics* 44, 4 (2025).
- Yang You, Mikaela Angelina Uy, Jiaqi Han, Rahul Thomas, Haotong Zhang, Yi Du, Hansheng Chen, Francis Engelmann, Suyu You, and Leonidas Guibas. 2025. Img2cad: Reverse engineering 3d cad models from images through vlm-assisted conditional factorization. In *Proceedings of the SIGGRAPH Asia 2025 Conference Papers*. 1–12.
- Zhanwei Zhang, Shizhao Sun, Wenxiao Wang, Deng Cai, and Jiang Bian. 2025. FlexCAD: Unified and Versatile Controllable CAD Generation with Fine-tuned Large Language Models. In *International Conference on Learning Representations (ICLR) 2025*. <https://arxiv.org/abs/2411.05823> arXiv:2411.05823.



Fig. 5. Here are some samples from our newly introduced dataset, PrintCAD, which comprises over 2,000 3D printed objects captured under uncontrolled real-world lighting conditions using an iPhone. The dataset aligns real-world images with corresponding ground-truth CAD models, spanning a variety of materials (Nylon, Resin, PLA) and geometric complexities.

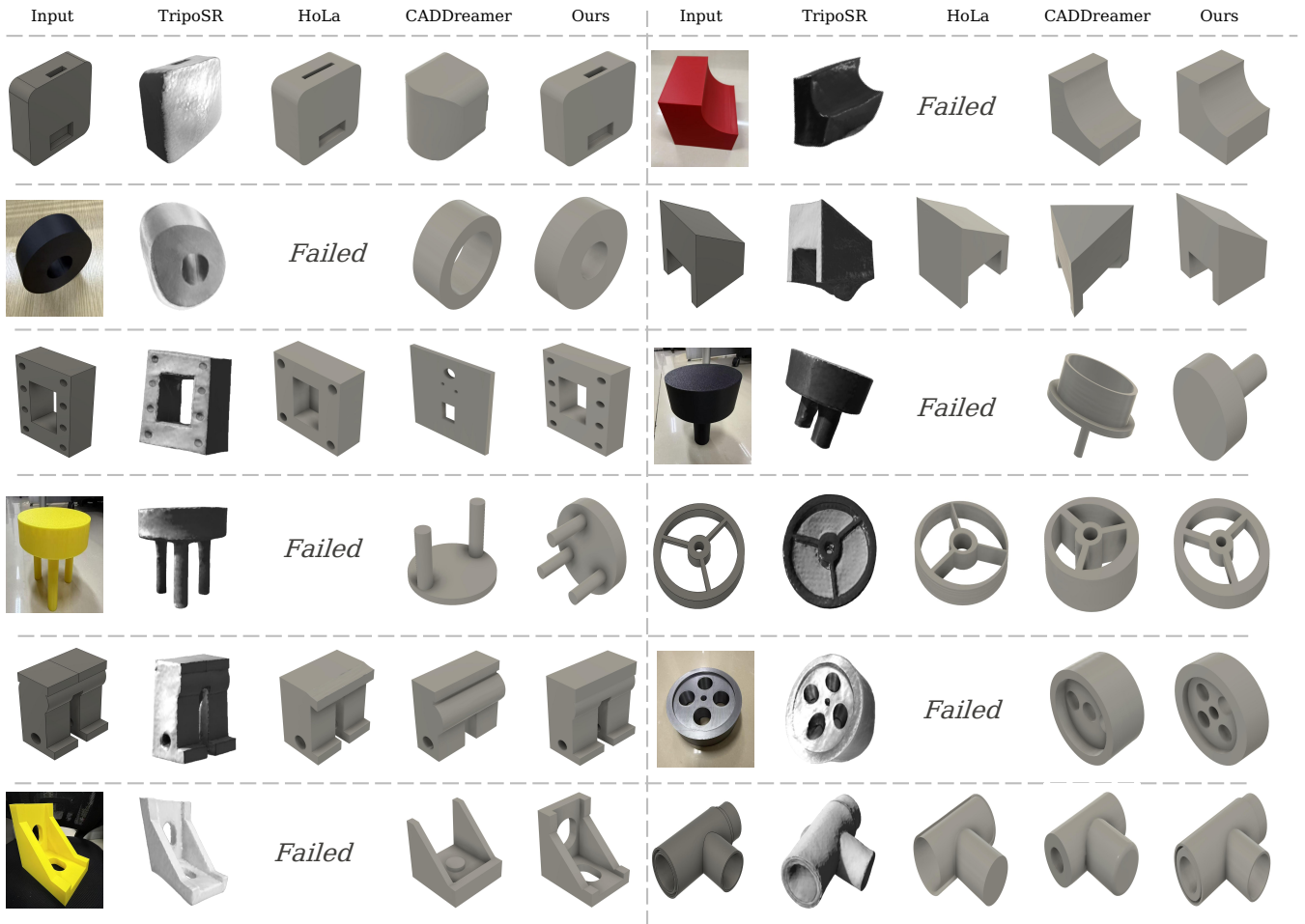


Fig. 6. We evaluate our method on synthetic and challenging real-world images. Since DeepCAD mostly contains simple shapes where all methods perform similarly, this figure highlights complex geometries to better illustrate our advantages. Unlike baselines that struggle with noise and smooth out sharp features, our approach reconstructs BRep models with rather cleaner topology and precise structural details.

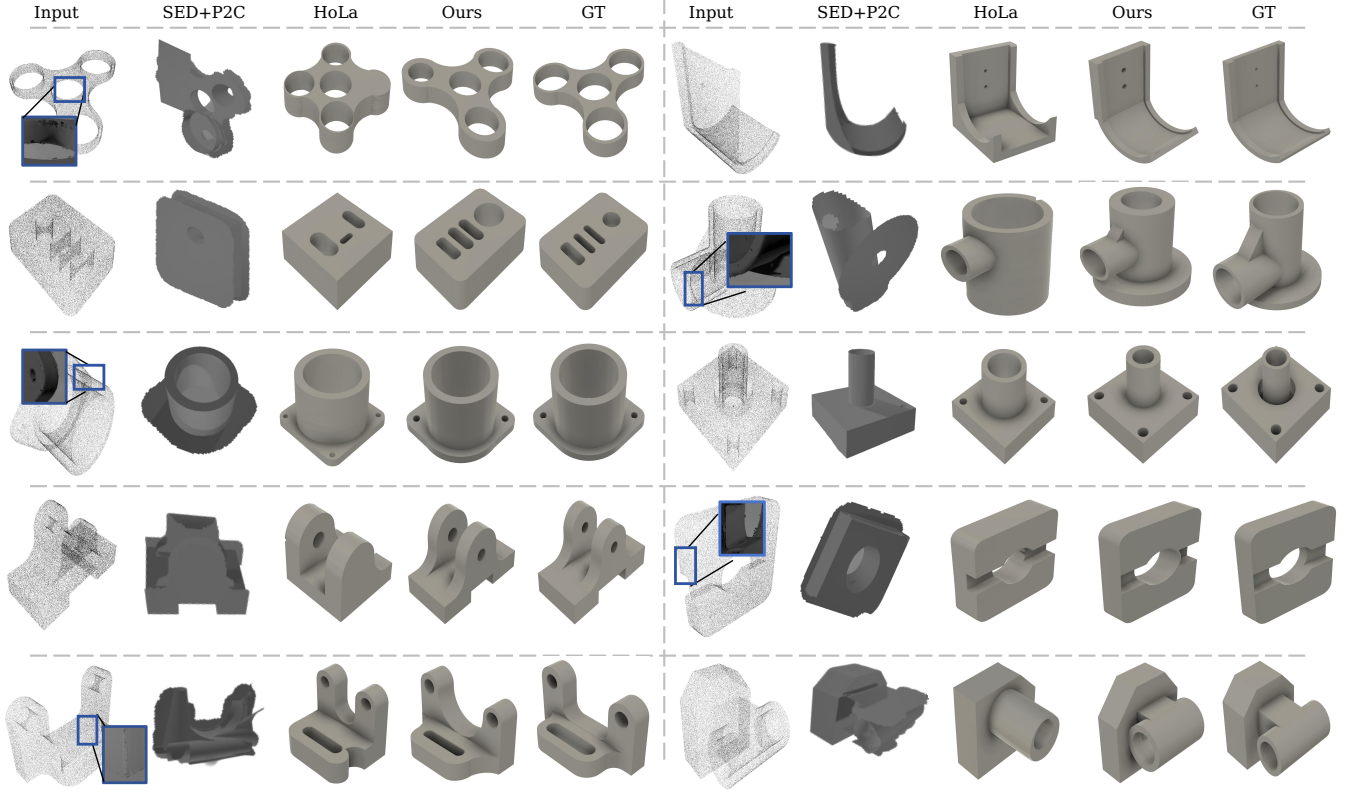


Fig. 7. We evaluate our method against state-of-the-art approaches on inputs with ill-scanned point clouds with misalignment parts, or the clean ones. While baseline methods often produce distorted shapes or fail to recover topology under heavy noise, our approach can still synthesize multiple plausible CAD candidates.

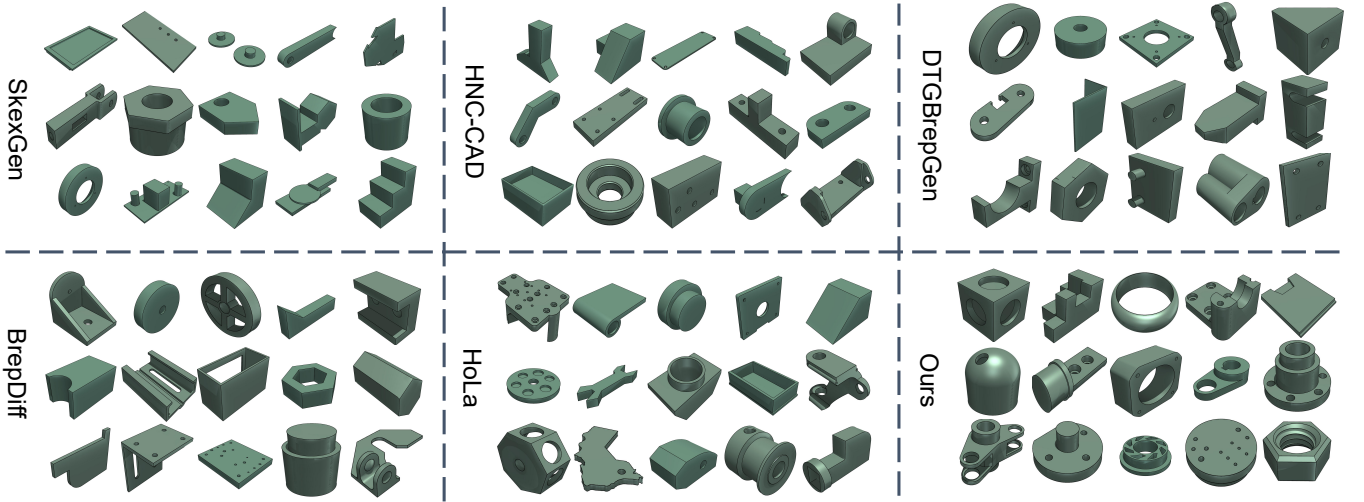


Fig. 8. We compare our Img2CADSeq with other widely adopted baselines in unconditional generation. Our method produces structurally plausible models with clean surfaces and coherent mechanical features, while baseline methods often exhibit simplistic shapes, poor assembly, or unnatural proportions, demonstrating our clear advantage in generating production-ready CAD candidates.